

Root Finding of Non-Linear Equations:

→ For powers of x which are low, we can easily find the root using approaches like:

eg $f(x) = x^2 - 2x = 0$

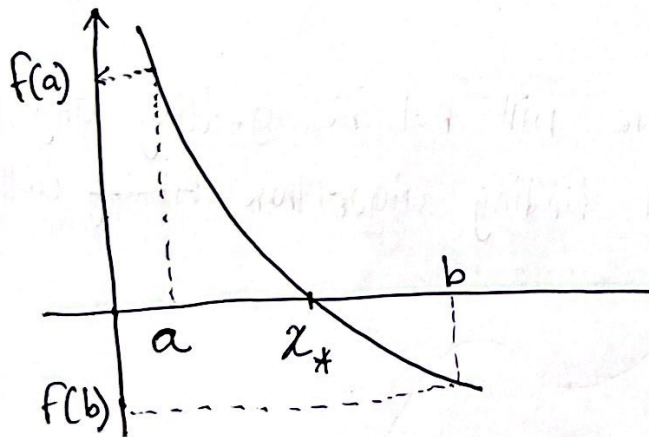
$$\Rightarrow x(x-2) = 0$$

$$x = 2, x = 0$$

→ But what if we are dealing with non-linear equations with high powers/series/polynomials/rational functions? How can we find their root?

↳ We can apply some iterative algorithms to find the roots.

Bisection Root Finding Algorithm:



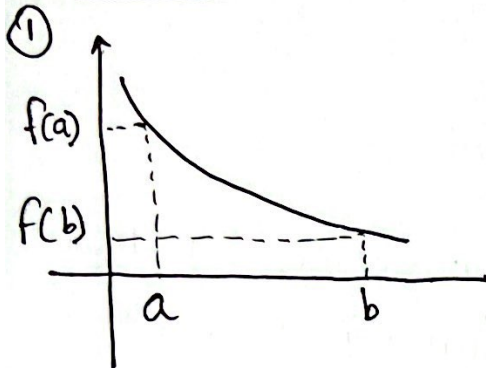
→ If root exists, the ~~sign~~ value of the function changes sign.

Here, $f(a) = +ve$
 $f(b) = -ve$

} Roots lies within $[a, b]$
~~changes~~ Therefore it changes sign

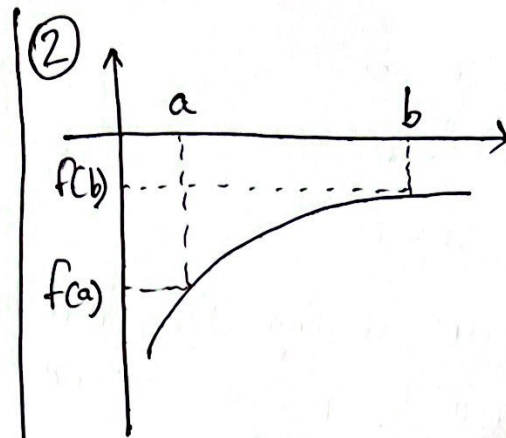
So, if $f(a) * f(b) < 0$ root exists between a and b .

Examples:



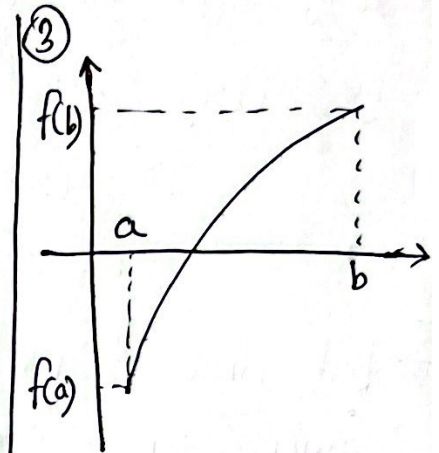
$$\begin{aligned} & f(a) * f(b) \\ &= (+ve) * (+ve) \\ &= (+ve) \\ &> 0 \end{aligned}$$

Root does not exist



$$\begin{aligned} & f(a) * f(b) \\ &= (-ve) * (-ve) \\ &= (+ve) \\ &> 0 \end{aligned}$$

Root does not exist

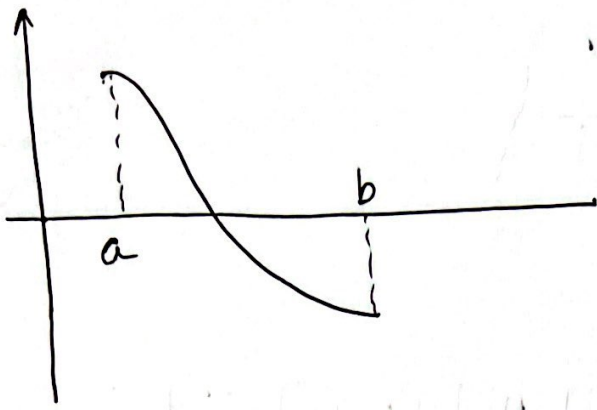


$$\begin{aligned} & f(a) * f(b) \\ &= (-ve) * (+ve) \\ &= (-ve) \\ &< 0 \end{aligned}$$

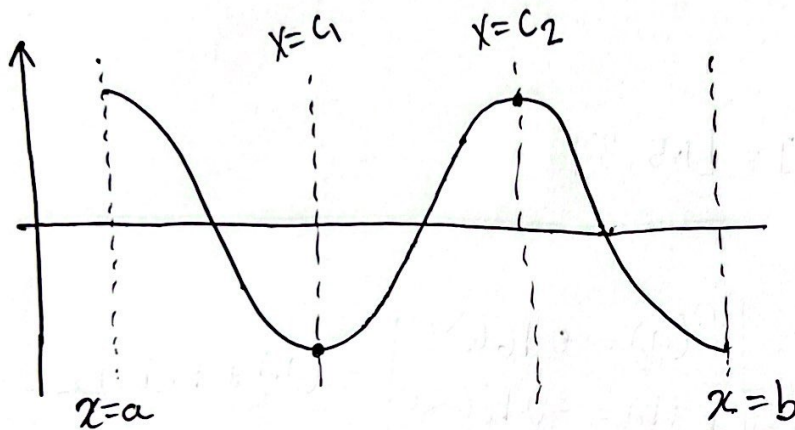
Root exists

For example ① and ②, we will not be getting any roots if we apply the any root finding algorithm within the interval $[a, b]$.

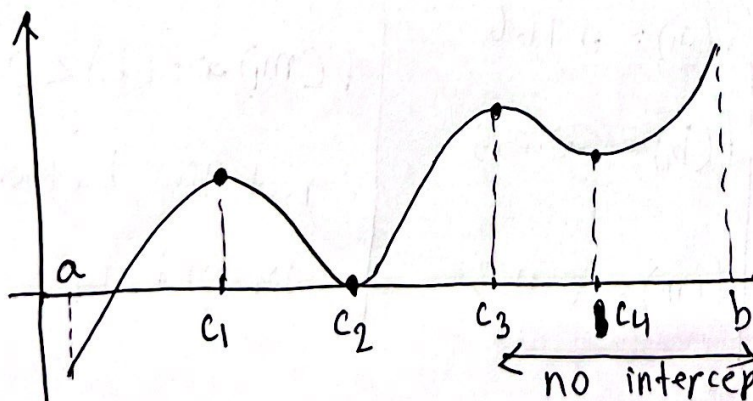
How to choose the right intervals:



$[a, b] \rightarrow$

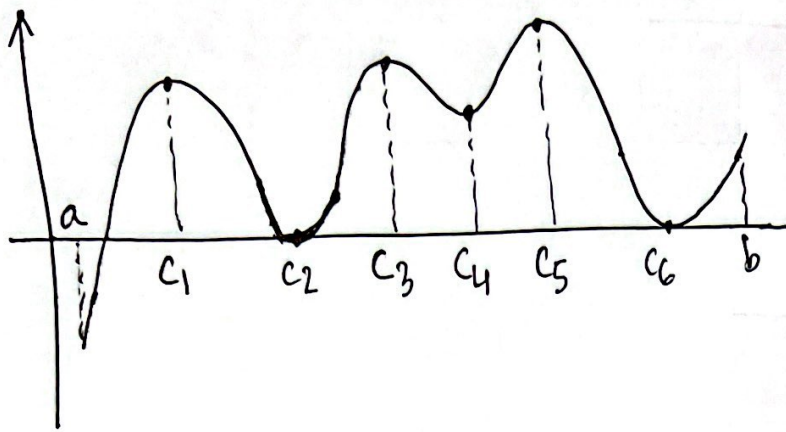


$$[a, b] = [a, c_1] \cup [c_1, c_2] \cup [c_2, b]$$



no intercept here, therefore can avoid.

$$[a, b] = [a, c_1] \cup [c_1, c_2] \cup [c_2, c_3]$$



$$[a, b] = [a, c_1] \cup [c_1, c_2] \cup [c_2, c_3] \cup [c_3, c_4] \cup [c_4, c_5] \cup [c_5, c_6] \cup [c_6, b]$$

Bisection Method:

$$f(x) = \frac{1}{x} - 0.5 \quad I = [1.5, 3]$$

$$a_0 = 1.5$$

$$b_0 = 3$$

$$m_0 = \frac{a_0 + b_0}{2} = \frac{1.5 + 3}{2} = 2.25$$

$$\left| \begin{array}{l} f(a_0) = 0.166 > 0 \\ f(b_0) = -0.166 < 0 \\ f(m_0) = -0.055 < 0 \end{array} \right|$$

$$f(a_0) * f(m_0) < 0$$

$\therefore$ root lies between a_0 and m_0

$$a_1 = a_0 = 1.5$$

$$b_1 = m_0 = 2.25$$

$$m_1 = \frac{a_1 + b_1}{2} = \frac{1.5 + 2.25}{2} = 1.875$$

$$\left| \begin{array}{l} f(a_1) = 0.166 \\ f(b_1) = -0.055 \\ f(m_1) = 0.033 \end{array} \right|$$

$$f(m_1) * f(b_1) < 0$$

$\therefore$ root lies between m_1 and b_1

Example

Use Bisection method to find solutions accurate within 10^{-3} for $f(x) = x^3 - 7x^2 + 14x - 6 = 0$ on interval $[1, 3.2]$

k	a_k	m_k	b_k	$f(a_k)$	$f(m_k)$	$f(b_k)$	$x_* \in [\quad , \quad]$
0	1	2.1	3.2	2	1.79	-0.11	$[2.1, 3.2]$
1	2.1	2.65	3.2	1.79	0.55	-0.11	$[2.65, 3.2]$
2	2.65	2.925	3.2	0.55	0.086	-0.11	$[2.925, 3.2]$
3	2.925	3.0625	3.2	0.86	-0.054	-0.11	$[2.925, 3.0625]$
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
9	2.998	3.000145	3.002	1.96×10^{-3}	-1.95×10^{-4} $< 10^{-3}$	-2.3×10^{-3}	

$$x_* = \underline{3.000}$$

Take upto 3 d.p. since error bound is 10^{-3} .

▣ Finding number of iterations to find root :

Things that will be provided in the question:

① interval, eg: $[1.5, 3]$

② Machine epsilon / error / accuracy, eg: 1.1×10^{-16}
bound

Formula:

$$n \geq \frac{\log(|b-a|) - \log(\epsilon)}{\log(2)} - 1$$

← put -1 if question specifies Machine epsilon or error. If question specifies accuracy, then do not put -1.

Example

interval = $[1.5, 3]$

$\epsilon_M = 1.1 \times 10^{-16}$

Find minimum amount of iteration required to find the root with the error bound of the machine epsilon.

Solution

$$n \geq \frac{\log|3-1.5| - \log(1.1 \times 10^{-16})}{\log(2)} - 1$$

$$\geq 52.59$$

$$\geq 53$$

Fixed Point Iteration:

→ We want to solve $f(x) = 0$

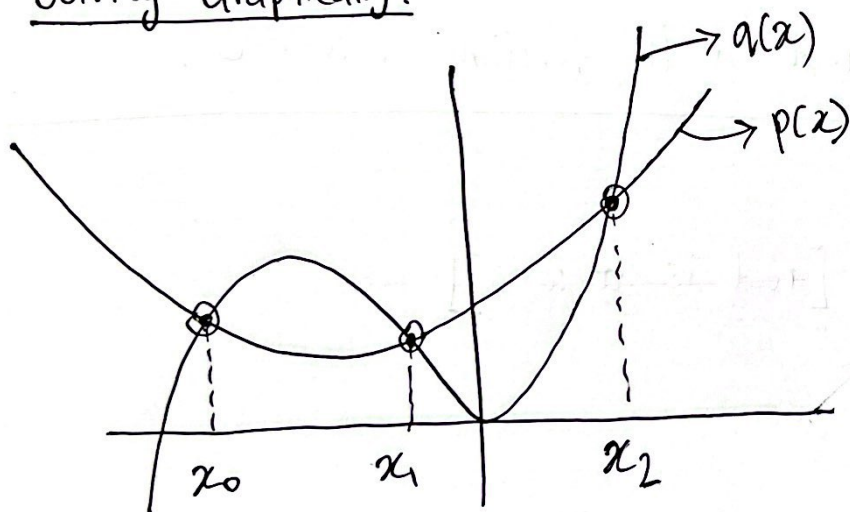
→ Need to transform $f(x) = 0$ into a new form

→ convert $f(x) = 0$ into $\boxed{g(x) = x}$

Algebra Recap:

$$\underbrace{x^2 + 5x + 7}_{p(x)} = \underbrace{x^3 + 3x^2}_{q(x)} \longrightarrow \underbrace{x^3 + 2x^2 - 5x - 7}_{f(x)}$$

Solving Graphically:

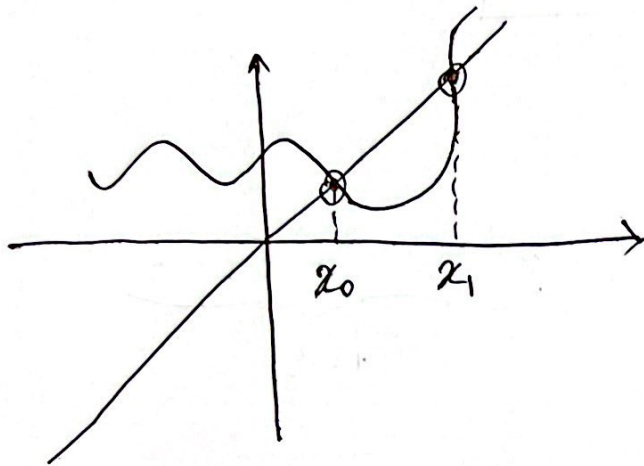


x_0, x_1, x_2 are the roots of the function $f(x)$

→ Now imagine we have a function $f(x) = 0$

→ Convert the function $f(x) = 0$ into $g(x) = x$

→ The intersection of $g(x) = x$ should give the root of the function $f(x) = 0$.



x_0, x_1 should be the root of the function $f(x) = 0$.

Example:

$$f(x) = -\frac{1}{2}x + 1 = 0 \quad [\text{root is at } x=2]$$

$$\underbrace{\frac{x+2}{2}}_{g(x)} = \underbrace{x}_x$$

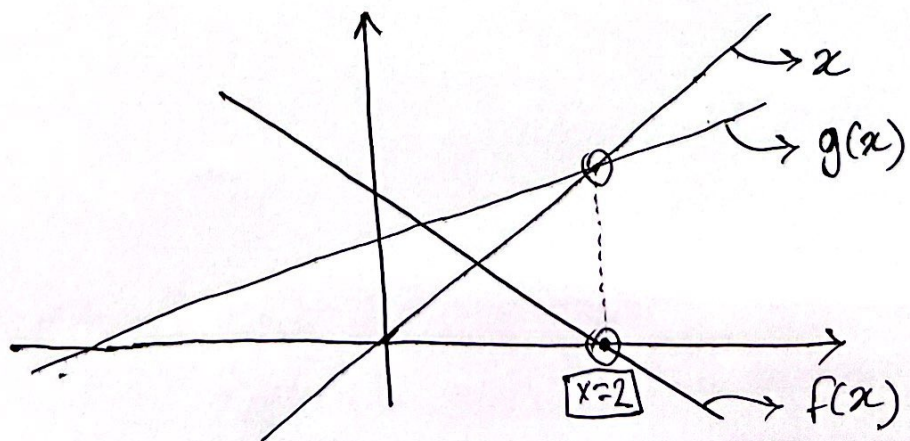
Solving numerically:

$$\frac{x+2}{2} = x$$

$$x+2 = 2x$$

$$\boxed{x=2}$$

Solving Graphically:



Converting $f(x)=0$ into $g(x)=x$:

$$f(x) = x^2 - 2x - 3 = 0$$

[Roots are at $x = -1, 3$]

[Found using middle term, or quadratic root finding formula]

make x the subject

$$\textcircled{1} \quad x^2 - 2x - 3 = 0$$

$$x^2 = 2x + 3$$

$$\underbrace{x}_{x} = \underbrace{\sqrt{2x+3}}_{g(x)}$$

$$\textcircled{2} \quad x^2 - 2x - 3 = 0$$

$$\Rightarrow x(x-2) - 3 = 0$$

$$\Rightarrow \underbrace{x}_{x} = \underbrace{\frac{3}{x-2}}_{g(x)}$$

$$\textcircled{3} \quad x^2 - 2x - 3 = 0$$

$$\Rightarrow x^2 - x - x - 3 = 0$$

$$\Rightarrow \underbrace{x}_{x} = \underbrace{\frac{x^2 - x - 3}{x}}_{g(x)}$$

$$\textcircled{4} \quad x^2 - 2x - 3 = 0$$

$$\Rightarrow 2x^2 - 2x = x^2 + 3$$

$$\Rightarrow x(2x-2) = x^2 + 3$$

$$\Rightarrow \underbrace{x}_{x} = \underbrace{\frac{x^2 + 3}{2x-2}}_{g(x)}$$

Root Finding Formula:

$$g(x_k) = x_{k+1}$$

① $g(x) = \sqrt{2x+3}$, $x_0 = 0$

$$g(0) = 1.73$$

$$g(1.73) = 2.54$$

$$g(2.54) = 2.84$$

$$g(2.84) = 2.95$$

$$g(2.95) = 2.98$$

$$g(2.98) = 2.99$$

$$g(2.99) = 3$$

$$g(3) = 3 \rightarrow \text{Fixed point reached.}$$

② $g(x) = x^2 - x - 3$, $x_0 = 0$

$$g(0) = -3$$

$$g(-3) = 9$$

$$\left. \begin{array}{l} g(9) = 69 \\ \vdots \end{array} \right\} \text{Diverges}$$

③ $g(x) = \frac{x^2+3}{2x-2}$, $x_0 = 0$

$$g(0) = -1.5$$

$$g(-1.5) = -1.05$$

$$g(-1.05) = -1$$

$$g(-1) = -1 \rightarrow \text{Fixed point reached.}$$

2 Questions to answer:

① At which root will it converge to?

→ It will depend on the initial choice of x_0

→ It will also depend on the value of the converging rate, λ .

② Which form of $g(x)$ is convergent?

→ It will depend on the value of the converging rate, λ .

Elaborating the first Q/A:

① $g(x) = \sqrt{2x+3}$, $x_0 = 0$ | $g(x) = \sqrt{2x+3}$, $x_0 = 42$

$$g(0) = 1.73$$

$$g(1.73) = 2.54$$

$$g(2.54) = 2.84$$

$$g(2.84) = 2.95$$

$$g(2.95) = 2.98$$

$$g(2.98) = 2.99$$

$$g(2.99) = 3.00$$

$$g(3) = 3$$

$$g(42) = 9.33$$

$$g(9.33) = 4.65$$

$$g(4.65) = 3.51$$

$$g(3.51) = 3.17$$

$$g(3.17) = 3.06$$

$$g(3.06) = 3.02$$

$$g(3.02) = 3.01$$

$$g(3.01) = 3$$

$$g(3) = 3$$

→ Both $x_0 = 0$ & $x_0 = 42$ converges to the root, $x_* = 3$.

→ Even though $x_0 = 0$ is closer to $x_* = -1$, it converges to $x_* = 3$.

$$\textcircled{2} \quad g(x) = x^2 - x - 3, \quad x_0 = 0 \quad \left| \quad g(x) = x^2 - x - 3, \quad x_0 = 42\right.$$

$$\begin{array}{l} g(0) = -3 \\ g(-3) = 9 \\ g(9) = 69 \\ g(69) = 4.69 \times 10^3 \\ \vdots \end{array} \quad \left. \vphantom{\begin{array}{l} g(0) = -3 \\ g(-3) = 9 \\ g(9) = 69 \\ g(69) = 4.69 \times 10^3 \\ \vdots \end{array}} \right\} \text{diverges}$$

$$\begin{array}{l} g(42) = 1.72 \times 10^3 \\ g(1.72 \times 10^3) = 2.95 \times 10^6 \\ g(2.95 \times 10^6) = 8.72 \times 10^{12} \\ \vdots \end{array} \quad \left. \vphantom{\begin{array}{l} g(42) = 1.72 \times 10^3 \\ g(1.72 \times 10^3) = 2.95 \times 10^6 \\ g(2.95 \times 10^6) = 8.72 \times 10^{12} \\ \vdots \end{array}} \right\} \text{Diverges}$$

→ Both $x_0 = 0$ and $x_0 = 42$ diverges rapidly.

$$\textcircled{3} \quad g(x) = \frac{x^2 + 3}{2x - 2}, \quad x_0 = 0 \quad \left| \quad g(x) = \frac{x^2 + 3}{2x - 2}, \quad x_0 = 42\right.$$

$$\begin{array}{l} g(0) = -1.5 \\ g(-1.5) = -1.05 \\ g(-1.05) = -1 \\ g(-1) = -1 \end{array} \quad \left. \vphantom{\begin{array}{l} g(0) = -1.5 \\ g(-1.5) = -1.05 \\ g(-1.05) = -1 \\ g(-1) = -1 \end{array}} \right\}$$

$$\begin{array}{l} g(42) = 21.6 \\ g(21.6) = 11.4 \\ g(11.4) = 6.39 \\ g(6.39) = 4.07 \\ g(4.07) = 3.19 \\ g(3.19) = 3.01 \\ g(3.01) = 3 \\ g(3) = 3 \end{array} \quad \left. \vphantom{\begin{array}{l} g(42) = 21.6 \\ g(21.6) = 11.4 \\ g(11.4) = 6.39 \\ g(6.39) = 4.07 \\ g(4.07) = 3.19 \\ g(3.19) = 3.01 \\ g(3.01) = 3 \\ g(3) = 3 \end{array}} \right\}$$

→ $x_0 = 0$ and $x_0 = 42$ converges to 2 different roots.

→ $x_0 = 0$ converges to the nearest root, which is -1 , and $x_0 = 42$ converges to the nearest root, which is 3 .
(42 is closer to 3 than -1).

Contraction Mapping Theory.

Find $g'(x)$

Find $g'(\text{root}-1)$

Find $g'(\text{root}-2)$

if $|g'(\text{root})| < 1$, $g(x)$ will be convergent.

$$\lambda = |g'(\text{root})|$$

↑
converging rate

$$\textcircled{1} \quad g(x) = \sqrt{2x+3} = (2x+3)^{\frac{1}{2}}$$

$$g'(x) = \frac{1}{2} (2x+3)^{-\frac{1}{2}} (2)$$

$$= (2x+3)^{-\frac{1}{2}}$$

$$= \frac{1}{\sqrt{2x+3}}$$

$$\lambda = |g'(-1)| = 1 \quad (\text{not } < 1)$$

$$\lambda = |g'(3)| = \frac{1}{3} \quad (< 1)$$

$\therefore$ both $x_0=0$, $x_0=42$ converges to the root $x_* = 3$.

$$\textcircled{2} \quad g(x) = x^2 - x - 3$$

$$g'(x) = 2x - 1$$

$$\lambda = |g'(-1)| = |-3| = 3 \quad (\text{not } < 1)$$

$$\lambda = |g'(3)| = 5 \quad (\text{not } < 1)$$

$\therefore$ no choices of x_0 will converge to any root. It will all diverge.

$$\textcircled{3} \quad g(x) = \frac{x^2 + 3}{2x - 2}$$

$$\lambda = |g'(-1)| = 0 \quad (< 1)$$

$$\lambda = |g'(3)| = 0 \quad (< 1)$$

$\therefore$ will converge to both $x_* = -1$ & 3 .

Since $x_0 = 0$ is closer to -1 , it will converge to -1 .

$x_0 = 42$ is closer to 3 , it will converge to 3 .

Example:

$$f(x) = x^3 - 2x^2 - x + 2$$

- (a) state the roots of the function $f(x)$
- (b) construct 3 different fixed point function $g(x)$ such that $f(x) = 0$
- (c) Find the converging rate of $g(x)$ and which root it will converge to.

Solution:

$$(a) f(x) = x^3 - 2x^2 - x + 2 = 0$$

$$\Rightarrow x^2(x-2) - 1(x-2) = 0$$

$$\Rightarrow (x^2 - 1)(x-2) = 0$$

$$x_* = +1$$

$$-1$$

$$+2$$

$$(b)(i) x^3 - 2x^2 - x + 2 = 0$$

$$x = \underbrace{x^3 - 2x^2 + 2}_{g(x)}$$

$$(ii) x^3 - 2x^2 - x + 2 = 0$$

$$\Rightarrow x(x^2 - 2x - 1) = -2$$

$$\Rightarrow x = \frac{-2}{x^2 - 2x - 1}$$

$$(iii) x^3 - 2x^2 - x + 2 = 0$$

$$\Rightarrow 2x^2 = x^3 - x + 2$$

$$\Rightarrow x^2 = \frac{1}{2}(x^3 - x + 2)$$

$$\Rightarrow x = \underbrace{\frac{1}{\sqrt{2}}}_{x} \underbrace{\sqrt{x^3 - x + 2}}_{g(x)}$$

$$(C)(i) \quad g(x) = x^3 - 2x^2 + 2$$

$$g'(x) = 3x^2 - 4x$$

$$\lambda = |g'(x_*)| = \left\{ \begin{array}{ll} 7 & \text{at } x_* = -1 \\ 1 & \text{at } x_* = 1 \\ 4 & \text{at } x_* = 2 \end{array} \right\} \text{ Divergent}$$

$$(ii) \quad g(x) = \frac{-2}{x^2 - 2x - 1}$$

$$g'(x) = \frac{4(x-1)}{(x^2 - 2x - 1)^2}$$

$$\lambda = |g'(x_*)| = \left\{ \begin{array}{ll} 2 & \text{at } x_* = -1 \\ 0 & \text{at } x_* = 1 \\ 4 & \text{at } x_* = 2 \end{array} \right.$$

$\therefore g(x)$ will converge to $x_* = 1$

$$(iii) \quad g(x) = \frac{1}{\sqrt{2}} (x^3 - x + 2)^{\frac{1}{2}}$$

$$g'(x) = \frac{3x^2 - 1}{2\sqrt{2} (x^3 - x + 2)^{1/2}}$$

$$\lambda = |g'(x_*)| = \left\{ \begin{array}{ll} 0.5 & \text{when } x_* = -1 \\ 0.5 & \text{when } x_* = 1 \\ 1.375 & \text{when } x_* = 2 \end{array} \right.$$

$\therefore g(x)$ will converge to $x_* = -1$ & 1 .

Order of convergence:

- ① $\lambda = 0 \rightarrow$ Super linear convergence $\rightarrow$ fastest convergence
 $\rightarrow$ less iterations required to converge.
- ② $0 < \lambda < 1 \rightarrow$ linear convergence $\rightarrow$ will converge
 $\rightarrow$ But not as fast as $\lambda = 0$.
- ③ $\lambda = 1 \rightarrow$ ignore
- ④ $\lambda > 1 \rightarrow$ Diverge .